

Scalability & Future Plan

Date	6 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection and learning Support Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Supports single-user processing at a time.	Reduced scalability under heavy user traffic.	High
2	Limited emotion detection to facial image analysis.	Cannot analyze voice or text-based emotions.	Medium
3	Learning recommendations are generated without long-term user history.	Reduced personalization over time.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited concurrent users.	Deploy load balancing and auto-scaling cloud infrastructure.
2	Data Storage	Minimal storage for application data.	Integrate a cloud database to store user profiles, emotion history, and recommendations.
3	Performance	AI processing depends on API response time.	Optimize API calls, implement caching, and use asynchronous processing.
4	Security	Basic authentication and API key protection.	Add JWT authentication, role-based access control, HTTPS enforcement, and encrypted data storage.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	User authentication and progress tracking.	1 week	Enables personalized learning experiences and secure user access.
Phase 3	Multi-language support and voice-based interaction.	10 days	Makes the application accessible to a wider range of users.
Phase 4	Learning analytics dashboard and LMS integration.	10 days	Provides detailed performance insights and seamless integration with educational platforms.